

In-Class Exercise 8 (solution)

*(i) Are the general equivalent positions in this group occupied and if so, by which atoms?
Are special positions occupied and if so, by which atoms?*

Atomic positions: V in (2c) $(1/4, 1/4, z)$ and $(3/4, 3/4, \bar{z})$
where $z = 0.84$

Mo in (2b) $(1/4, 3/4, 1/2)$ and $(3/4, 1/4, 1/2)$

O(1) in (2c) $(1/4, 1/4, z)$ and $(3/4, 3/4, \bar{z})$
where $z = 0.233$

O(2) in (8g) $(x, y, z); (1/2 - x, 1/2 - y, z); (1/2 - y, x, z); (y, 1/2 - x, z);$
 $(\bar{x}, \bar{y}, \bar{z}); (1/2 + x, 1/2 + y, \bar{z}); (1/2 + y, \bar{x}, \bar{z});$
 $(\bar{y}, 1/2 + x, \bar{z})$
where $x = 0.703, y = 0.462, z = 0.26$

- General position $\rightarrow$ lowest symmetry and highest multiplicity; we need to know the order of the group
- In $P4/n$, the tetrad axis along z creates 4 points and the diagonal glide plane in the x - y plane creates 4 more $\rightarrow$ total of **8 general positions**
- So, **8g** sites have the correct multiplicity to be general
- Remainder of atoms are in lower multiplicity special positions!

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(ii) Draw a diagram of the structure projected onto the (001) plane. Show all of the atoms in one unit cell, labeling the atom type and the vertical position.

Table 3.4. The crystal structure of VOMoO_4 [21]

Formula unit	VOMoO_4
Space group:	$P4/n$ (no. 85)
Cell dimensions:	$a = 6.608 \text{ \AA}$, $c = 4.265 \text{ \AA}$
Cell contents:	2 formula units
Atomic positions:	V in (2c) $(1/4, 1/4, z)$ and $(3/4, 3/4, \bar{z})$ where $z = 0.84$
	Mo in (2b) $(1/4, 3/4, 1/2)$ and $(3/4, 1/4, 1/2)$
	O(1) in (2c) $(1/4, 1/4, z)$ and $(3/4, 3/4, \bar{z})$ where $z = 0.233$
	O(2) in (8g) (x, y, z) ; $(1/2 - x, 1/2 - y, z)$; $(1/2 - y, x, z)$; $(y, 1/2 - x, z)$; $(\bar{x}, \bar{y}, \bar{z})$; $(1/2 + x, 1/2 + y, \bar{z})$; $(1/2 + y, \bar{x}, \bar{z})$; $(\bar{y}, 1/2 + x, \bar{z})$ where $x = 0.703$, $y = 0.462$, $z = 0.26$

Tetragonal structure →
(001) projection is a square



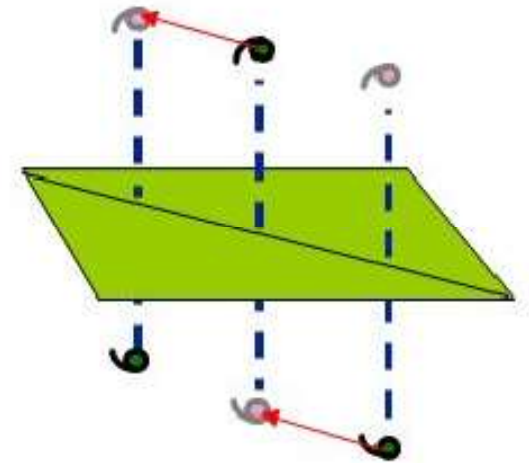
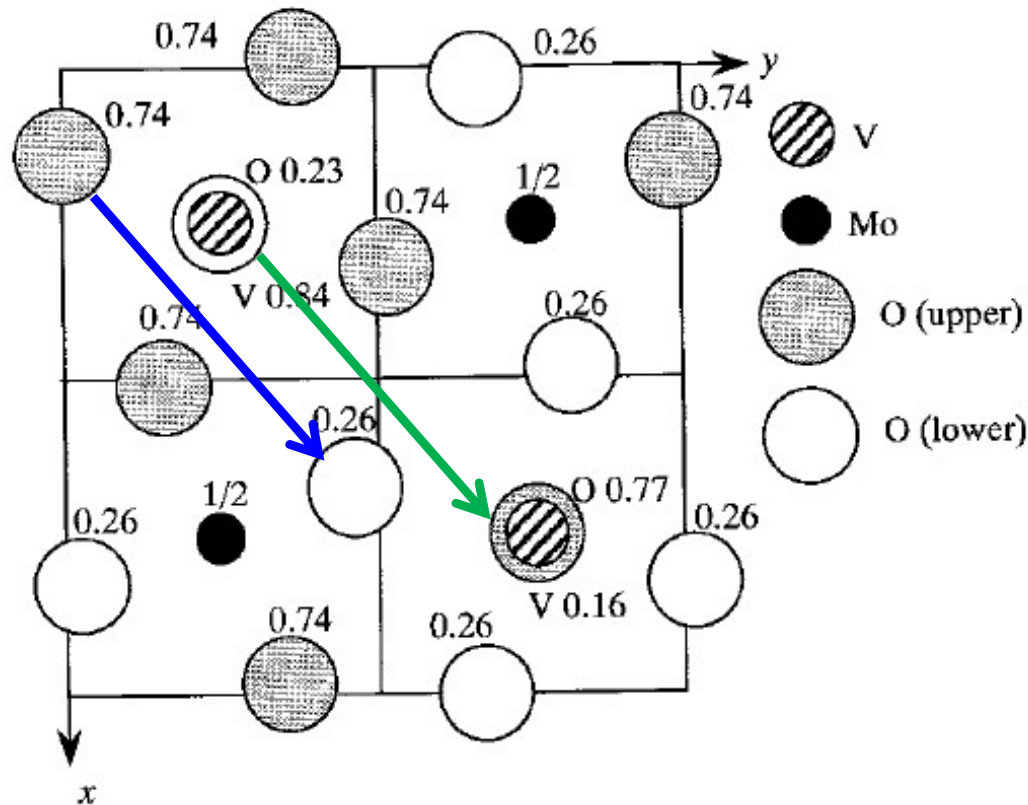
$$a = b \neq c$$

$$\alpha = \beta = \gamma = 90^\circ$$

Coordinates:

V	$(1/4, 1/4, 0.84)$	$(3/4, 3/4, 0.16)$		
Mo	$(1/4, 3/4, 1/2)$	$(3/4, 1/4, 1/2)$		
O1	$(1/4, 1/4, 0.233)$	$(3/4, 3/4, 0.767)$		
O2	$(0.703, 0.462, 0.26)$	$(0.797, 0.038, 0.26)$	$(0.038, 0.703, 0.26)$	$(0.462, 0.797, 0.26)$
	$(0.297, 0.538, 0.74)$	$(0.203, 0.962, 0.74)$	$(0.962, 0.297, 0.74)$	$(0.536, 0.203, 0.74)$

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(iii) Where is the n operator in this structure?

From the symbol, we know the n glide plane is $\perp$ tetrad and therefore lies in the x - y plane ($4/n$); it takes points from (x, y, z) , to $(x+1/2, y+1/2, \bar{z})$.

Based on the atomic positions of upper and lower O atoms and V atoms in the above diagram, the glide planes are situated at $z = 0$ (and 1) and $z = 1/2$.

This is consistent with the Mo atoms remaining fixed at $z=1/2$